Joel Castillo Rey

PhD Student

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Education

2023–Present **PhD Researcher**, *Basque Center for Applied Mathematics (BCAM)*, Bilbao, Spain
Supervisor: Ilya Smirnov.

Fall 2022 **Research Internship**, *BCAM*, Bilbao, Spain

2021–2022 **Master's Degree in Advanced Mathematics**, *Universidad Complutense de Madrid*, Madrid, Spain

2017–2021 **Degree in Mathematics**, *Universidad Complutense de Madrid*, Madrid, Spain

Research interests

Commutative Algebra, Algebraic Geometry and Singularity Theory. Local Algebra in Positive Characteristic.

Publications

2026 ***F-invariants and essential deformations***, *with B. Mudrak, preprint, to appear on arXiv*

2026 ***Limit F-invariants of the simple singularities and the Fermat hypersurfaces***, *preprint, arXiv*

2025 ***Strong Watanabe–Yoshida conjecture for complete intersections***, *preprint, arXiv*

Invited Talks

Sep 2026 **EPFL**, *Facets of Frobenius derived invariants (Bernoulli Center)*, Lausanne, Switzerland

Mar 2026 **University of the Basque Country**, *Algebra Seminar*, Bilbao, Spain

Nov 2025 **University of Michigan**, *CA Seminar*, Michigan, USA

Nov 2025 **University of Illinois in Chicago**, *AG Seminar*, Illinois, USA

Oct 2025 **Purdue University**, *CA Seminar*, Indiana, USA

Jan 2024 **Universidad de Sevilla**, *RGAS School on Singularities*, Seville, Spain

Nov 2023 **ICMAT**, *5th BYMAT Conference*, Madrid, Spain

Sep 2022 **Universidad Complutense de Madrid**, *XVI Workshop de Jóvenes Investigadores*, Madrid, Spain

Visits

Aug–Dec 2025 **Research Stay**, *Purdue University*, Indiana, USA
Host: Uli Walther

Posters

Jul 2026 **CAAG in Milan**, Milan, Italy
Linkage and the Watanabe–Yoshida conjecture.

Jun 2025 **Singularities in Algebra and Geometry Conference (CIMAT)**, Guanajuato, Mexico
The Strong Watanabe–Yoshida conjecture for complete intersections.

A selection of other activities

Oct 2026 **Simons School on Higher Dimensional Varieties**, *Rényi Institute*, Budapest, Hungary

Oct 2025 **KUMUNU 2025**, *University of Missouri*, Columbia, USA

Nov 2024 **Sheaf Cohomology Course**, Zaragoza, Spain

Apr 2023 **BCAM-Severo Ochoa Course: Introduction to multiplicity theory**, *BCAM*, Bilbao, Spain